MATH MA’AM ACADEMY

Where Every Subject Clicks.

STD 5 MATHEMATICS – CHAPTER 6: PERIMETER AND AREA

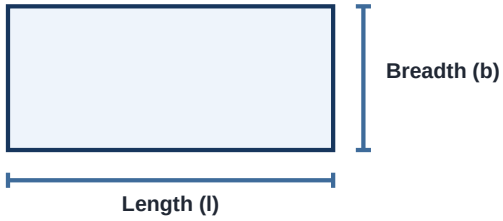
Learning objectives: Understand perimeter and area; calculate perimeter of rectangles, squares and triangles; calculate area of rectangles and squares; estimate area by counting unit squares.

6.1 PERIMETER

Perimeter is the total distance around the boundary or outer edges of a closed figure. To find the perimeter, add the lengths of all its sides.

Perimeter is a length. It is measured in units such as mm, cm, m and km — never in square units.

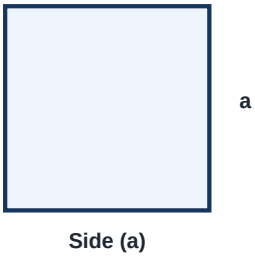
Perimeter of common shapes



Rectangle: Opposite sides of a rectangle are equal. It has two lengths (l) and two breadths (b).

$$\text{Perimeter} = l + b + l + b = 2 \times (l + b)$$

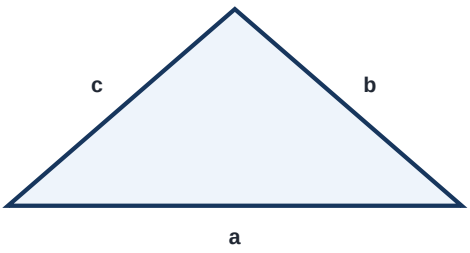
Example: Length = 12 cm, breadth = 5 cm.
Perimeter = 2 × (12 + 5) = 2 × 17 = **34 cm**.



Square: All four sides of a square are equal. If each side is a, then:

$$\text{Perimeter} = a + a + a + a = 4 \times a$$

Example: Side = 8 cm. Perimeter = 4 × 8 = **32 cm**.



Triangle: A triangle has three sides. Add all three side lengths to find its perimeter.

$$\text{Perimeter} = a + b + c$$

Example: Sides = 5 cm, 7 cm and 9 cm. Perimeter = 5 + 7 + 9 = **21 cm**.

Solving perimeter word problems

• **Fencing a field:** Find the perimeter first. If fencing is required twice around, multiply the perimeter by 2.

• **Ribbon around a photo frame:** Add the lengths of all four sides.

• **Finding a missing side:** Subtract the known side lengths from the total perimeter.

• **Same perimeter:** Different shapes can have the same perimeter but different areas.

Worked example: A rectangular garden is 18 m long and 7 m wide. Fencing needed for one round = 2 × (18 + 7) = 50 m.

6.2 AREA

Area is the amount of surface or space covered inside a flat, two-dimensional (2D) shape.

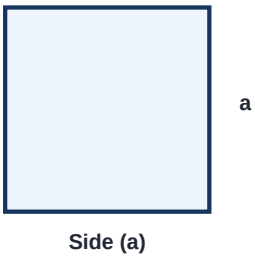
Area is measured in square units such as mm², cm² and m². Area is not measured in cm or m.

Area of common shapes

Rectangle: Multiply its length by its breadth.

$$\text{Area} = \text{Length} \times \text{Breadth} = l \times b \text{ square units}$$

Example: Length = 9 cm, breadth = 4 cm. Area = 9 × 4 = **36 cm²**.



Square: Multiply the side by itself.

$$\text{Area} = \text{Side} \times \text{Side} = a \times a = a^2 \text{ square units}$$

Example: Side = 7 cm. Area = 7 × 7 = **49 cm²**.

Finding a missing dimension

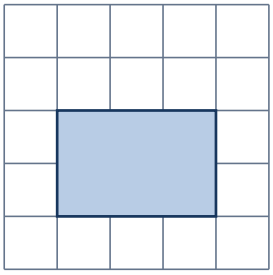
Given	To find	Method
Area and length of a rectangle	Breadth	Breadth = Area ÷ Length
Area and breadth of a rectangle	Length	Length = Area ÷ Breadth
Area of a square	Side	Side × Side = Area; find the number which multiplied by itself gives the a

Example: A rectangle has area 72 cm² and length 9 cm. Breadth = 72 ÷ 9 = **8 cm**.

Example: A square has area 100 cm². Its side is 10 cm because 10 × 10 = 100.

Area by counting unit squares

A square of side 1 cm has an area of 1 cm². On a square grid, count the complete shaded squares and combine partial squares to estimate the area.



Each small square
= 1 square unit

Part of a square	Area counted using the estimation rule
One complete shaded square	1 square unit
More than half of a square is shaded	1 square unit
Exactly half of a square is shaded	½ square unit
Less than half of a square is shaded	0 square units

Worked example: A figure contains 2 complete shaded squares, 2 more-than-half shaded squares, 4 half-shaded squares and 2 less-than-half shaded squares.

Estimated area = (2 × 1) + (2 × 1) + (4 × ½) + (2 × 0)
= 2 + 2 + 2 + 0 = 6 square units.

Perimeter and area – do not confuse them

Perimeter	Area
Distance around the boundary	Space covered inside the shape
Add the side lengths	Use the relevant area formula
Units: cm, m, km	Square units: cm², m², km²
A one-dimensional measurement	A two-dimensional measurement

Olympiad tips and common mistakes

• Read the question carefully: does it ask for the boundary distance or the surface covered?

• For a rectangle, do not multiply length and breadth when the question asks for perimeter.

• For a square, perimeter is 4 × side, while area is side × side.

• Write the correct unit in every final answer: cm for perimeter and cm² for area.

• For a missing rectangle side, divide area by the known side; do not subtract.

• For a grid figure, count whole and partial squares according to the stated rule.

Quick formula recap
Rectangle perimeter = 2 × (l + b)
Square perimeter = 4 × a
Triangle perimeter = a + b + c
Rectangle area = l × b
Square area = a × a

ALL THE BEST